

of solar power. When it comes to roof-top solar installations, people are concerned about the high upfront costs involved and do not want to risk taking the first step.

Are you really aware about the investments to go solar and the commercials of it? Let us find the answers here.

Drastic reduction in costs of solar PV panels

Solar energy is generated through solar photovoltaic (PV) panels that are installed on rooftops. Major cost involved in any solar system is the cost of these solar panels. In early 1977, per watt cost of these panels was US\$ 76.67 as shown in Fig. 1. Over a period of time, this price dropped considerably to US\$ 0.36 in 2014.

Although the graph in Fig. 1 represents the fall in prices (popularly called Swanson Effect), prices may vary as per the make (brand) of a particular solar panel. Typical price per watt of a good Indian-make solar panel can vary from ₹ 32 to ₹ 37, whereas for a solar panel from Tata Power the price can go up to ₹ 45 per watt. The price is further expected to fall over the next two years.

Note. The rooftop solar systems that are being talked of here are grid-tied and grid-interactive type where solar power thus generated can be transferred to the grid.

Cost varies with size

The government's initiative to increase installation of solar systems by allowing the sale of energy generated has created a commercial sense of solar power. This commer-

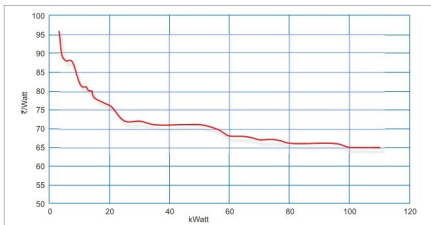


Fig. 2: Typical system cost per watt of solar (data gathered by Sunkalp)

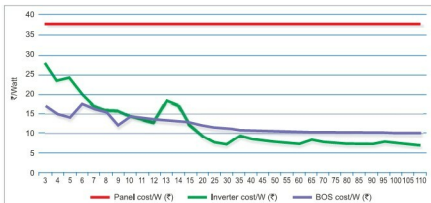


Fig. 3: Cost per watt for each component (data gathered by Sunkalp)

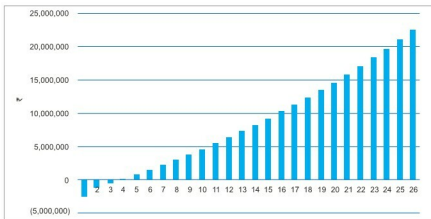


Fig. 4: Cumulative cash flow for a 40kW system



For More
Test and Equipment Stories
Visit www.electronicstest2b.com